Numerical Optimization 15: Constrained Optimization

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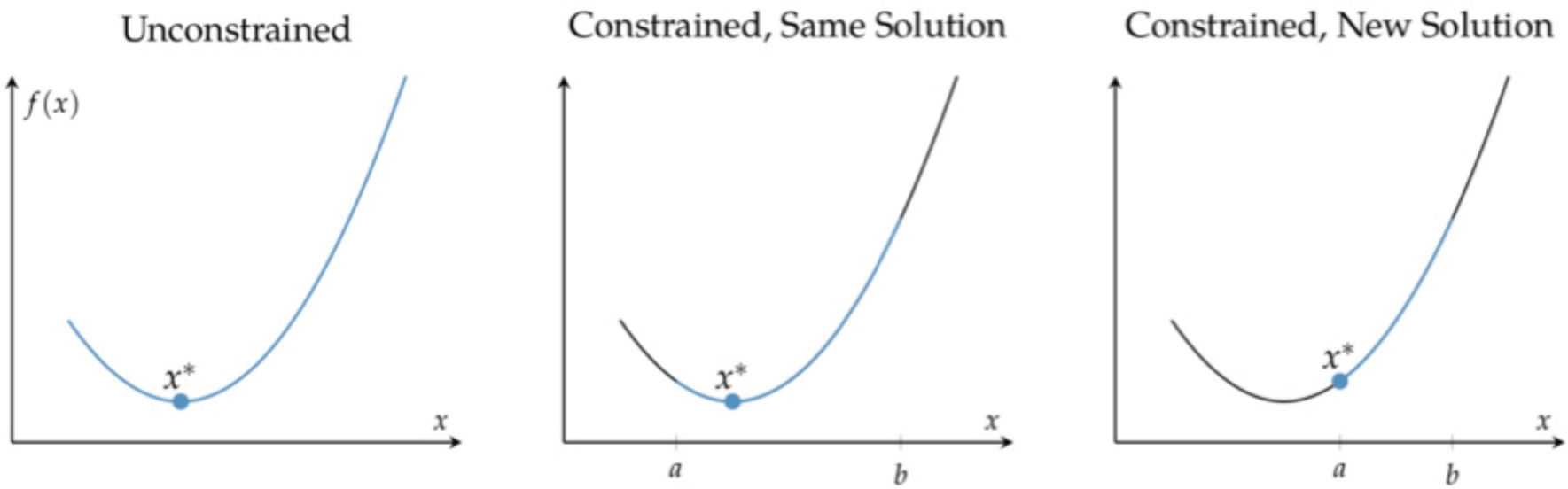
May 17, 2020

Overview

1. Constrained Optimization
2. Constraints
3. Transformations to Remove Constraints
4. Lagrange Multipliers
5. Summary

Noisy Descent

Fitting Surrogate Models



Constraints

Constraints are not typically specified directly through a known feasible set X . Instead, the feasible set is typically formed from two types of constraints:

- equality constraints, $h(x) = 0$
- inequality constraints, $g(x) \leq 0$

Any optimization problem can be rewritten using these constraints

$$\begin{aligned} & \min_{\boldsymbol{x}} \ f(\boldsymbol{x}) \\ s.t. \quad & h_i(x) = 0 \\ & g_j(x) = 0 \end{aligned}$$

Transformations to Remove Constraints

In some cases, it may be possible to transform a problem so that constraints can be removed. For example, bound constraints a x b can be removed by passing x through a transform

$$x = \frac{b+a}{2} + \frac{b-a}{2}\left(\frac{2\hat{x}}{1+\hat{x}^2}\right)$$

Below is an example

$$\begin{aligned} &\min_{x} \; x \sin x \\ &s.t. \quad 2 \leq x \leq 6 \end{aligned}$$

Can be transformed to

$$\min_{\hat{x}} \left[4 + 2\left(\frac{2\hat{x}}{1+\hat{x}^2}\right)x + \sin\left[4 + 2\frac{2\hat{x}}{1+\hat{x}^2}\right]\right.$$

Lagrange Multipliers

The method of Lagrange multipliers is used to optimize a function subject to equality constraints.

$$\min_{\boldsymbol{x}} f(\boldsymbol{x})$$
$$s.t. \quad h_i(x) = 0$$

where f and h have continuous partial derivatives.
We can formulate the Lagrangian, which is a function of the design variables,

$$\mathcal{L}(x, \lambda) = f(x) - \lambda h(x)$$

Solving $\nabla\mathcal{L}(x, \lambda) = 0$. Specifically, $\nabla_x\mathcal{L} = 0$ gives us the condition $\nabla f = \lambda\nabla h$, and $\nabla\lambda\mathcal{L} = 0$ gives us $h(x) = 0$. Any solution is considered a critical point.

Lagrange Multipliers to a single equality condition

The method of Lagrange multipliers is used to optimize a function subject to equality constraints.

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & -\exp[-(x_1x_2 - 3/2)^2 - (x_2 - 3/2)^2] \\ s.t. \quad & x_1 - x_2^2 = 0 \end{aligned}$$

We can formulate the Lagrangian,

$$\mathcal{L}(x, \lambda) = -\exp[-(x_1x_2 - 3/2)^2 - (x_2 - 3/2)^2] + \lambda(x_1 - x_2^2)$$

We compute

- $\frac{\partial \mathcal{L}}{\partial x_1}$
- $\frac{\partial \mathcal{L}}{\partial x_2}$
- $\frac{\partial \mathcal{L}}{\partial \lambda}$

Lagrange Multipliers to multiple equality conditions

The method of Lagrange multipliers is used to optimize a function subject to equality constraints.

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & -\exp[-(x_1x_2 - 3/2)^2 - (x_2 - 3/2)^2] \\ s.t. \quad & x_1 - x_2^2 = 0 \end{aligned}$$

We can formulate the Lagrangian,

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We compute

- $\frac{\partial \mathcal{L}}{\partial x_1}$
- $\frac{\partial \mathcal{L}}{\partial x_2}$
- $\frac{\partial \mathcal{L}}{\partial \lambda}$

Summary

- Constraints are requirements on the design points that a solution must satisfy.
- Some constraints can be transformed or substituted into the problem to result in an unconstrained optimization problem.
- Analytical methods using Lagrange multipliers yield the generalized Lagrangian and the necessary conditions for optimality under constraints.
- A constrained optimization problem has a dual problem formulation that is easier to solve and whose solution is a lower bound of the solution to the original problem.